

UNIT-5: TEXT MINING

Text Mining: Text mining is a component of data mining that deals specifically with unstructured text data. It involves the use of natural language processing (NLP) techniques to extract useful information and insights from large amounts of unstructured text data. Text mining can be used as a preprocessing step for data mining or as a standalone process for specific tasks. Text mining in data mining is mostly used for, the unstructured text data that can be transformed into structured data that can be used for data mining tasks such as classification, clustering, and association rule mining. This allows organizations to gain insights from a wide range of data sources, such as customer feedback, social media posts, and news articles.

Text Mining vs. Text Analytics

Text mining and text analytics are related but distinct processes for extracting insights from textual data. Text mining involves the application of natural language processing and machine learning techniques to discover patterns, trends, and knowledge from large volumes of unstructured text. However, Text Analytics focuses on extracting meaningful information, sentiments, and context from text, often using statistical and linguistic methods. While text mining emphasizes uncovering hidden patterns, text analytics emphasizes deriving actionable insights for decision-making. Both play crucial roles in transforming unstructured text into valuable knowledge, with text mining exploring patterns and text analytics providing interpretative context.

Areas of text mining:

These are the following area of text mining:

- **InformationExtraction:**

The automatic extraction of structured data such as entities, entities

relationships, and attributes describing entities from an unstructured source is called information extraction.

- **Natural Language Processing:**

NLP stands for Natural language processing. Computer software can understand human language as same as it is spoken. NLP is primarily a component of artificial intelligence (AI). The development of the NLP application is difficult because computers generally expect humans to "Speak" to them in a programming language that is accurate, clear, and exceptionally structured. Human speech is usually not authentic so that it can depend on many complex variables, including slang, social context, and regional dialects.

- **Data Mining:**

Data mining refers to the extraction of useful data, hidden patterns from large data sets. Data mining tools can predict behaviors and future trends that allow businesses to make a better data-driven decision. Data mining tools can be used to resolve many business problems that have traditionally been too time-consuming.

- **Information Retrieval:**

Information retrieval deals with retrieving useful data from data that is stored in our systems. Alternately, as an analogy, we can view search engines that happen on website retrieval.

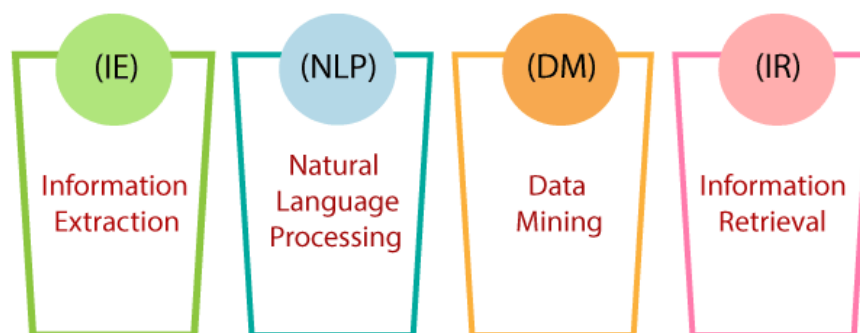


Fig: Area's of Text Mining

Text Mining Process Overview:

The text mining process incorporates the following steps to extract the data from the document.

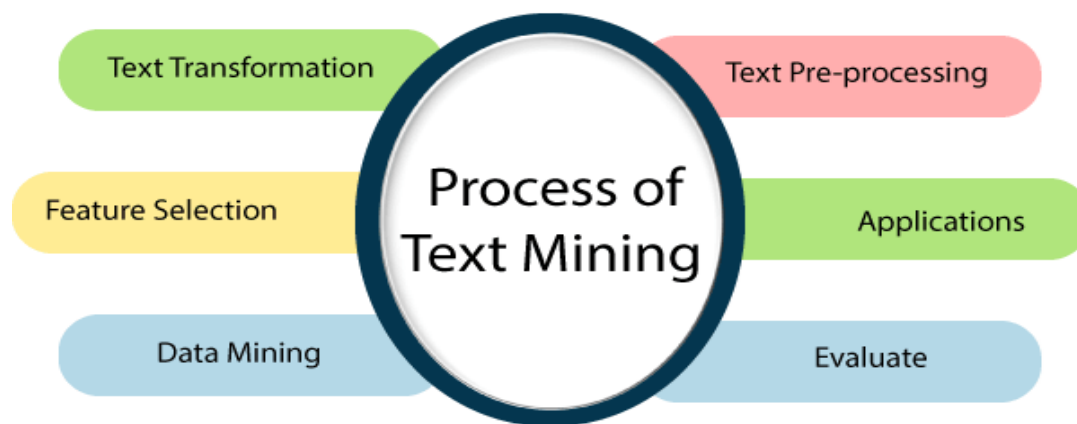


Fig: Process of Text Mining

- **TextTransformation:**

A text transformation is a technique that is used to control the capitalization of the text. Here the two major way of document representation is given.

- a. Bag of words
- b. Vector Space

- **TextPre-processing:**

Pre-processing is a significant task and a critical step in Text Mining, Natural Language Processing (NLP), and information retrieval (IR). In the field of text mining, data pre-processing is used for extracting useful information and knowledge from unstructured text data. Information Retrieval (IR) is a matter of

choosing which documents in a collection should be retrieved to fulfill the user's need.

- **FeatureSelection:**

Feature selection is a significant part of data mining. Feature selection can be defined as the process of reducing the input of processing or finding the essential information sources. The feature selection is also called variable selection.

- **DataMining:**

Now, in this step, the text mining procedure merges with the conventional process. Classic Data Mining procedures are used in the structural database.

- **Evaluate:**

Afterward, it evaluates the results. Once the result is evaluated, the result abandon.

- **Applications:**

These are the following text mining applications:

- i) RiskManagement:**

Risk Management is a systematic and logical procedure of analyzing, identifying, treating, and monitoring the risks involved in any action or process in organizations. Insufficient risk analysis is usually a leading cause of disappointment. It is particularly true in the financial organizations where adoption of Risk Management Software based on text mining technology can effectively enhance the ability to diminish risk. It enables the administration of millions of sources and petabytes of text documents, and giving the ability to connect the data. It helps to access the appropriate data at the right time.

- ii) CustomerCareService:** Text mining methods, particularly NLP, are finding increasing significance in the field of customer care. Organizations are spending in text analytics programming to improve their overall experience by accessing the

textual data from different sources such as customer feedback, surveys, customer calls, etc. The primary objective of text analysis is to reduce the response time of the organizations and help to address the complaints of the customer rapidly and productively.

iii)BusinessIntelligence:

Companies and business firms have started to use text mining strategies as a major aspect of their business intelligence. Besides providing significant insights into customer behavior and trends, text mining strategies also support organizations to analyze the qualities and weaknesses of their opponent's so, giving them a competitive advantage in the market.

iv)SocialMediaAnalysis:

Social media analysis helps to track the online data, and there are numerous text mining tools designed particularly for performance analysis of social media sites. These tools help to monitor and interpret the text generated via the internet from the news, emails, blogs, etc. Text mining tools can precisely analyze the total no of posts, followers, and total no of likes of our brand on a social media platform that enables us to understand the response of the individuals who are interacting with our brand and content.

Importance of Text Mining

Text mining is a powerful technique for extracting valuable insights from large volumes of textual data. Its importance spans multiple domains:

- **Knowledge discovery:** Helps uncover hidden patterns, trends, and relationships in textual information, aiding decision-making.
- **Customer insights:** Businesses use text mining to analyze customer feedback, reviews, and social media interactions to improve products and services.

- **Healthcare and biomedical research:** Extracts meaningful information from medical records, research papers, and clinical notes, supporting diagnosis and treatment advancements.
- **Fraud detection and security:** Identifies fraudulent transactions, cybersecurity threats, and suspicious activities by analyzing textual patterns.
- **Sentiment analysis:** Determines public opinion, market sentiment, and consumer preferences by processing social media, news, and review data.
- **Legal and Compliance:** Helps legal professionals sift through massive volumes of case laws, contracts, and regulatory documents efficiently.
- **Academic Research:** Assists in literature reviews, identifying key references, and tracking academic trends across various fields.

Advantages of Text Mining

- **Large Amounts of Data:** Text mining allows organizations to extract insights from large amounts of unstructured text data.
- **Variety of Applications:** Text mining has a wide range of applications, including sentiment analysis, named entity recognition, and topic modeling.
- **Improved Decision Making**
- **Cost-effective:** Text mining can be a cost-effective way, as it eliminates the need for manual data entry.

Disadvantages of Text Mining

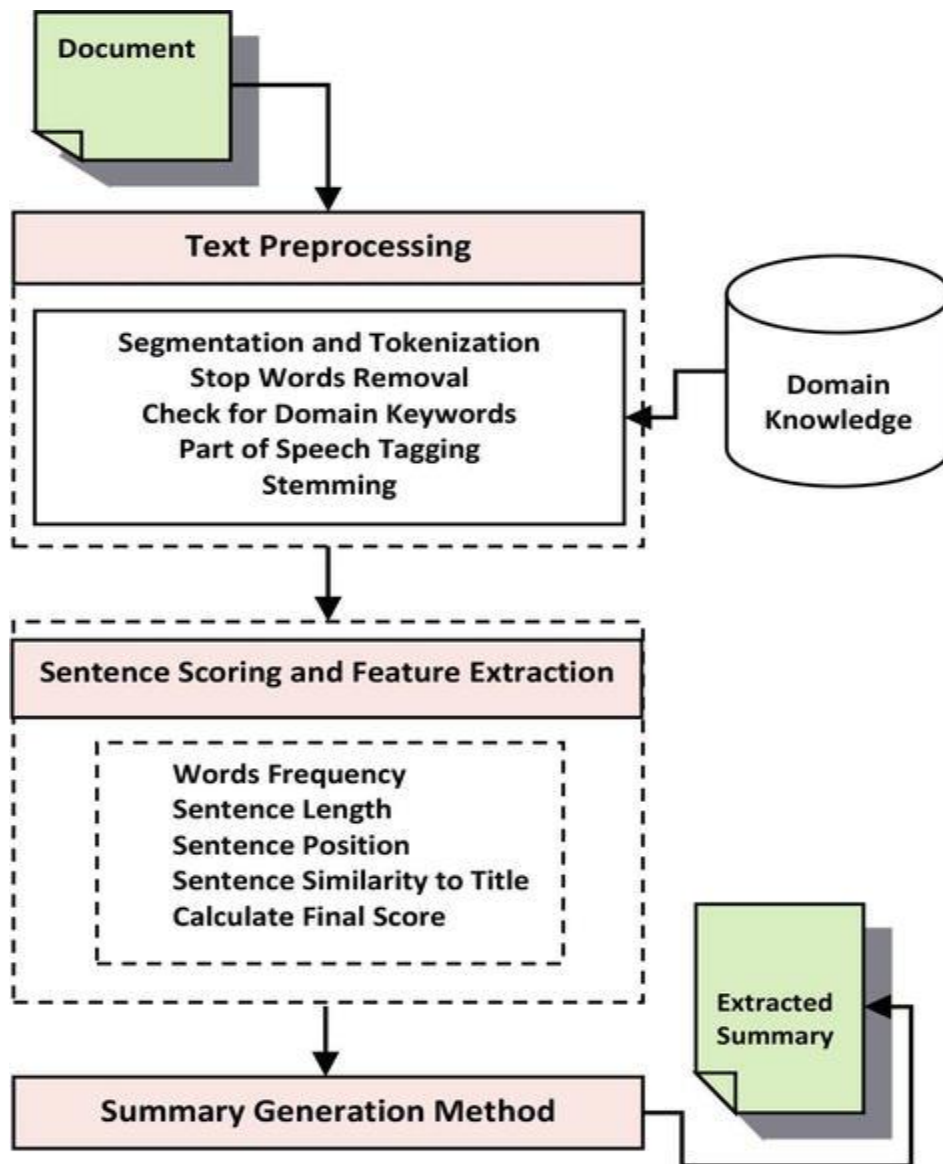
- **Complexity:** Text mining can be a complex process requiring advanced skills in natural language processing and machine learning.
- **Quality of Data:** The quality of text data can vary, affecting the accuracy of the insights extracted from text mining.

- **High Computational Cost:** Text mining requires high computational resources, and it may be difficult for smaller organizations to afford the technology.
- **Limited to Text Data:** Text mining is limited to extracting insights from unstructured text data and cannot be used with other data types.
- **Noise in text mining results:** Text mining of documents may result in mistakes. It's possible to find false links or to miss others. In most situations, if the noise (error rate) is sufficiently low, the benefits of automation exceed the chance of a larger mistake than that produced by a human reader.
- **Lack of transparency:** Text mining is frequently viewed as a mysterious process where large corpora of text documents are input and new information is produced. Text mining is in fact opaque when researchers lack the technical know-how or expertise to comprehend how it operates, or when they lack access to corpora or text mining tools.

Text Preprocessing:

Text preprocessing is the process of cleaning and converting text data into a usable format. It is a necessary step before using any text mining techniques. Common text preprocessing techniques includes:

- i) **Tokenization:** breaking up text into separate tokens, which can be individual words, phrases, or whole sentences.
- ii) **Term frequency:** counting the number of times a term appears in a document.
- iii) **Stemming:** reducing words to their root form.
- iv) **Lemmatization:** reducing words to their base form.



Difference between data mining and text mining

S.No.	Data Mining	Text Mining
1.	Data mining is the statistical technique of processing raw data in a structured form.	Text mining is the part of data mining which involves processing of text from documents.

S.No.	Data Mining	Text Mining
2.	Pre-existing databases and spreadsheets are used to gather information.	The text is used to gather high quality information.
3.	Processing of data is done directly.	Processing of data is done linguistically.
4.	Statistical techniques are used to evaluate data.	Computational linguistic principles are used to evaluate text.
5.	In data mining data is stored in structured format.	In text mining data is stored in unstructured format.
6.	Data is homogeneous and is easy to retrieve.	Data is heterogeneous and is not so easy to retrieve.
7.	It supports mining of mixed data.	In text mining, mining of text is only done.
8.	It combines artificial intelligence, machine learning and statistics and applies it on data.	It applies pattern recognizing and natural language processing to unstructured data.

S.No.	Data Mining	Text Mining
9.	It is used in fields like marketing, medicine, healthcare.	It is used in fields like bioscience and customer profile analysis.

Text Clustering

- Text clustering is the application of cluster analysis to text-based documents.
- It uses machine learning and natural language processing (NLP) to understand and categorize unstructured, textual data.
- Text clustering can be used anywhere from product development to customer support.
- Text clustering is a process of grouping most similar articles, tweets, reviews, and documents together. Here each group is known as a cluster.
- In clustering, documents within-cluster are similar and documents in different clusters are dissimilar. There are various clustering techniques available such as K-Means, DBSCAN, Spectral clustering, and hierarchical clustering.
- Clustering is known as the data segmentation method. It partitions the large data sets into similar groups. Clustering can also be utilized in outlier detection problems such as fraud detection and monitoring of criminal activities.
- Text Clustering is a broadly used unsupervised technique in text analytics. Text clustering has various applications such as clustering or organizing documents and text summarization. Clustering is also used in various applications such as customer segmentation, recommender system, and visualization. Text mining or analytics techniques need text to be converted into some type of vectors such as

Bag of Words(BoW), Term Frequency-Inverse Document Frequency (TF-IDF), Word2Vec, Doc2Vec, Sent2Vec, USE, Skip-thoughts, or other transformers.

Text Similarity:

Text similarity is the process of comparing two pieces of text and determining how similar they are. It's a fundamental task in natural language processing (NLP) that helps us understand the degree of closeness between texts. Text Similarity is the process of comparing a piece of text with another and finding the similarity between them. It's basically about determining the degree of closeness of the text. Dealing with text, sentences or words brings us to the region of ***Natural Language Processing (NLP)***, where we are going to use different NLP approaches to process the raw text and help the model to detect the similarity more swiftly and efficiently.

Text Assemble in Text Mining:

Text assembly in text mining involves gathering, organizing, and preprocessing textual data to create a structured and analyzable dataset. This process is essential for extracting meaningful insights from unstructured text data. Here's a detailed overview of the steps involved in text assembly:

1. Data Collection:

- **Sources:** Identify and gather text data from various sources such as websites, social media, emails, research papers, books, reports, etc.
- **Tools and Methods:** Use web scraping tools (e.g., BeautifulSoup, Scrapy), APIs, or manual collection methods.

2. Data Cleaning:

- **Removing Noise:** Eliminate irrelevant information such as HTML tags, advertisements, and navigation links.
- **Dealing with Missing Data:** Address incomplete data by either filling in missing values, ignoring those records, or using imputation techniques.

- **Standardization:** Convert text to a consistent format, such as lowercase, to ensure uniformity.

3. Text Preprocessing:

- **Tokenization:** Split text into individual words or phrases (tokens).
- **Stop Word Removal:** Remove common words (e.g., "and," "the," "is") that do not contribute much to the meaning.
- **Stemming and Lemmatization:** Reduce words to their base or root form (e.g., "running" to "run").
- **Normalization:** Handle variations in words, such as different spellings (e.g., "color" vs. "colour").

4. Feature Extraction:

- **Bag of Words (BOW):** Represent text as a collection of words, ignoring grammar and word order but keeping multiplicity.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Reflect how important a word is to a document in a corpus.
- **Word Embedding:** Use techniques like Word2Vec or GloVe to capture semantic meaning by representing words in continuous vector space.

5. Text Representation:

- **Document-Term Matrix (DTM):** Create a matrix where rows represent documents and columns represent terms, with cell values indicating term frequency.
- **N-grams:** Capture sequences of n words to preserve some context (e.g., bigrams, trigrams).

6. Data Integration:

- **Combining Data Sources:** Integrate data from different sources to enrich the dataset.
- **Metadata Inclusion:** Add metadata such as author, date, and source to provide additional context.

7. Data Transformation:

- **Dimensionality Reduction:** Use techniques like Principal Component Analysis (PCA) to reduce the number of features while retaining important information.
- **Feature Engineering:** Create new features that can better represent the underlying patterns in the text data.

8. Handling Imbalanced Data:

- **Resampling Techniques:** Use oversampling or under sampling methods to balance the dataset.
- **Synthetic Data Generation:** Create synthetic examples using methods like SMOTE (Synthetic Minority Over-sampling Technique).

9. Data Formatting:

- **Structured Data:** Convert the text data into a structured format (e.g., CSV, JSON) suitable for analysis.
- **Storage:** Store the processed data in databases or data lakes for easy retrieval and analysis.

10. Validation and Quality Assurance:

- **Quality Checks:** Ensure the accuracy and completeness of the assembled data.
- **Consistency:** Verify that the preprocessing steps have been consistently applied across the dataset.

By following these steps, we can effectively assemble text data, making it ready for various text mining techniques such as sentiment analysis, topic modeling, and named entity recognition.

Text Exploration:

Text exploration in text mining involves analyzing and understanding the characteristics, patterns, and structures within a text dataset. This process helps in gaining insights and identifying significant trends, themes, or anomalies before applying more complex text mining techniques. Here are the key steps involved in text exploration:

1. Initial data inspection:

- **Overview:** Start by examining the dataset to understand its size, structure, and content. This can include checking the number of documents, average length of documents, and basic statistics.
- **Metadata Analysis:** Look at metadata such as author, date, source, and any other contextual information.

2. Basic Statistics:

- **Word Count:** Calculate the total number of words, unique words, average words per document, etc.
- **Frequency Distribution:** Analyze the frequency of individual words and phrases to identify the most common terms.
- **Document Length Distribution:** Plot the distribution of document lengths to understand the variability.

3. Visualization Techniques:

- **Word Clouds:** Create word clouds to visualize the most frequent terms in the dataset. Larger words in the cloud represent higher frequency.

- **Bar Plots:** Plot the frequencies of the top N words or phrases using bar charts.
- **Histograms:** Use histograms to visualize the distribution of various textual features like word count, sentence length, etc.
- **Term Co-occurrence Matrix:** Visualize the relationships between terms using heat maps or network graphs.

4. Lexical Diversity:

- **Vocabulary Richness:** Measure the richness of the vocabulary using metrics like Type-Token Ratio (TTR), which is the ratio of unique words (types) to total words (tokens).
- **Entropy:** Calculate the entropy to understand the unpredictability or randomness in the text data.

5. Text Structure Analysis:

- **Sentence Structure:** Analyze the average sentence length, sentence complexity, and the distribution of various parts of speech (POS) such as nouns, verbs, adjectives, etc.
- **Paragraph Structure:** Examine paragraph lengths and structure to understand document composition.

6. Contextual Analysis:

- **Concordance Analysis:** Look at how specific words or phrases are used in context by generating concordance lines.
- **N-grams Analysis:** Identify and analyze common bigrams, trigrams, and higher-order n-grams to capture word sequences and patterns.

7. Sentiment Analysis:

- **Basic Sentiment Scores:** Apply basic sentiment analysis techniques to get an overview of the emotional tone of the documents.

- **Sentiment Distribution:** Plot the distribution of sentiment scores to understand the overall sentiment landscape.

8. Topic Identification:

- **Keyword Extraction:** Use techniques like TF-IDF to extract significant keywords from the text.
- **Topic Modeling:** Apply preliminary topic modeling methods (e.g., Latent Dirichlet Allocation - LDA) to identify potential topics within the text.

9. Named Entity Recognition (NER):

- **Entity Extraction:** Identify and extract named entities such as persons, organizations, locations, dates, and other relevant entities.
- **Entity Frequency:** Analyze the frequency and distribution of different entities in the dataset.

10. Trends and Patterns:

- **Temporal Analysis:** If the dataset includes time-based metadata, analyze trends over time, such as changes in word usage, sentiment, or topics.
- **Geospatial Analysis:** If location data is available, explore geographic trends and patterns.

11. Interactive Exploration:

- **Interactive Tools:** Use interactive text exploration tools (e.g., Jupyter notebooks with Pandas and Plotly, or specialized tools like Orange or KNIME) to dynamically explore and visualize the data.

12. Clustering and Similarity:

- **Document Clustering:** Apply clustering techniques(e.g.,K-means, hierarchical clustering) to group similar documents together.
- **Similarity Analysis:** Measure similarities between documents using cosine similarity or other distance metrics to identify closely related texts.